



About this overview. This document is an example of how the course might be taught. More topics and activities are approved for this course than appear here, so scholars may see some variation in topics and activities from course to course, and between the live school and On-Demand. Each teacher also leans into their own strengths and passions; the structure, expectations, and time commitment below are representative of the course as designed.

ABOUT THIS COURSE

An NGSS + CPALMS-aligned course covering the practice of science, matter and its properties, chemical and physical changes, force and motion, energy, Earth systems, space, ecosystems, the human body, and engineering design. Scholars design investigations, build and revise models, analyze data, and write evidence-based explanations. The course culminates in a capstone where scholars independently apply the year's science practices.

THE YEAR AT A GLANCE

U1 — Practice of Science & Investigation

Observation, testable questions, variables, fair tests.

U2 — Matter & Its Properties

Particles, states, measuring mass/volume.

U3 — Chemical & Physical Changes

Mixtures, solutions, signs of chemical change.

U4 — Force & Motion

Balanced/unbalanced forces, gravity, friction.

U5 — Energy & Its Forms

Electrical circuits, heat transfer, sound, light.

U6 — Earth's Systems: Water & Weather

Water cycle, weather patterns, climate.

U7 — Earth's Place in the Universe

Sun-Earth-Moon system, stars, seasons.

U8 — Ecosystems & Interactions

Food webs, interdependence, human impact.

U9 — The Human Body

Organ systems, structures and functions.

U10 — Engineering Design & Capstone

Design process, testing, year-end capstone.

Capstone & projects. The final modules close the year with a capstone project where scholars independently apply investigation design, modeling, and evidence-based reasoning.

A TYPICAL WEEK

- **Tuesday:** Core content lesson — vocabulary, models, data interpretation, CER practice.
- **Thursday:** Investigation, lab, model-building, engineering challenge, or data task with evidence-based explanation.

TIME COMMITMENT

160 minutes per week (master schedule)

- 2 modules ● Content lesson: ~80 min on Tuesday
- 3 modules ● Investigation / lab: ~80 min on Thursday
- 3 modules ● Includes embedded written explanations and quizzes.
- Engineering and capstone weeks may run longer.

WHAT SCHOLARS NEED

- Science notebook or OneDrive
- Simple lab materials for home investigations
- Microsoft 365 / OneDrive account
- VR headset for virtual investigations on Optima's VR campus

GRADING AT A GLANCE

- Assignments — 60%
- Quizzes — 15%
- Summative Assessments — 25%

On-Demand vs. Live. On-Demand (asynchronous): self-paced with due dates for pacing; a teacher is available for support, feedback, and grading. **Live school:** live-school scholars take the same course on Optima's VR campus with a teacher and classmates (on the school's weekly schedule).